

Bookstore Mini Inventory Management System

Programming Exercise / Problem Statement

Duration	1.5 hours
Java Stream	Core Java is fine. However, feel free to use Spring or JEE.
RoR Stream	Please use Ruby and not RoR.

Problem Statement

A bookstore needs a mini inventory management system with the following facilities:

1. Create Inventory

Inventory is just a collection of various items. Each item can have these details: ID, name, and cost.

2. Update Item Stock

Inventory is updated whenever a new stock is received. For example, when the inventory has 5 pens, we say that there are 5 pens in stock. When a new stock of 5 pens arrives, the inventory is updated to have a total of 10 pens in stock.

3. Checkout Item

When an item is being sold, it first needs to be checked out from the inventory. For example, when the store has 10 pens in stock and a customer asks for 3 pens, checking out those 3 pens reduces the stock of pens from 10 to 7.

4. Generate Inventory Report

Generate a simple report of all the items in the inventory. The report should indicate the ID, name, cost, and stock (count of items) available for each item.

Implement this mini inventory management system.

Notes

1. It is desirable to have a library facilitating the above operations as part of the implementation. Imagine a case where this library can be provided as a JAR to be used in some backend application.
2. Data need not be persisted into a database or a file system. Hold all the data in memory; losing data on application shutdown is okay.
3. Concurrency handling is a plus. Imagine a case where inventory replenishment and inventory checkout are done at the same time. Managing correct inventory becomes crucial.
4. Handle corner cases appropriately. Good error handling and logging are desirable. Use logging facilities such as `log4j` or `java.util.logging`.
5. Beautiful code is appreciated!